



Lecture # 11 & 12

Project Quality Management

ENGINEERING PROJECT MANAGEMENT

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What is Project Quality?

- ▶ The International Organization for Standardization (ISO) defines **quality** as “the degree to which a set of inherent characteristics fulfils requirements” (ISO9000:2000)
- ▶ Other experts define quality based on:
 - ▶ **Conformance to requirements** : The project's processes and products meet written specifications
 - ▶ **Fitness for use**: A product can be used as it was intended

Project Quality Management - Processes

- ▶ **Project quality management** ensures that the project will satisfy the needs for which it was undertaken
- ▶ Processes include:
 - 1.Planning quality management:** Identifying which quality standards are relevant to the project and how to satisfy them; a **metric** is a standard of measurement
 - 2.Performing quality assurance:** Periodically evaluating overall project performance to ensure the project will satisfy the relevant quality standards
 - 3.Performing quality control:** Monitoring specific project results to ensure that they comply with the relevant quality standards

Project Quality Management - Summary

Planning

Process: **Plan quality management**

Outputs: Quality management plan, process improvement plan, quality metrics, quality checklists, and project documents updates

Executing

Process: **Perform quality assurance**

Outputs: Change requests, project management plan updates, project documents updates, and organizational process asset updates

Monitoring and Controlling

Process: **Perform quality control**

Outputs: Quality control measurements, validated changes, validated deliverables, work performance information, change requests, project management plan updates, project documents updates, and organizational process asset updates

Project Start

Project Finish

Controlling Quality

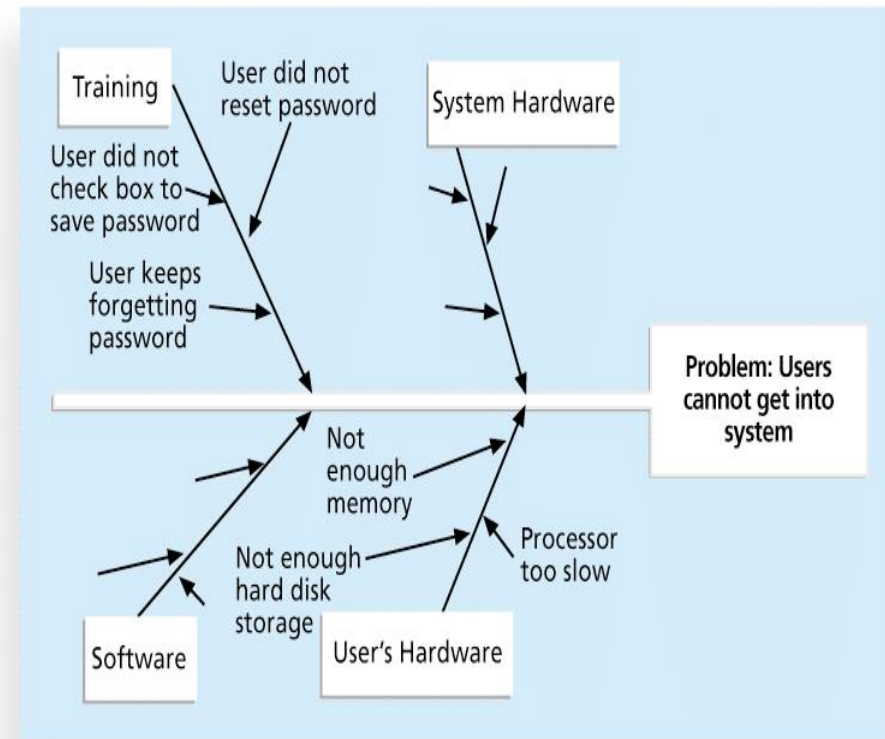
- ▶ The main outputs of quality control are:
 - ▶ Acceptance decisions
 - ▶ Rework
 - ▶ Process adjustments
- ▶ There are Seven Basic Tools of Quality that help in performing quality control

Controlling Quality

- ▶ Cause and effect diagrams
- ▶ Quality control charts – the seven run rule
- ▶ Check sheet
- ▶ Scatter diagram
- ▶ Histograms
- ▶ Pareto charts
- ▶ Flow charts

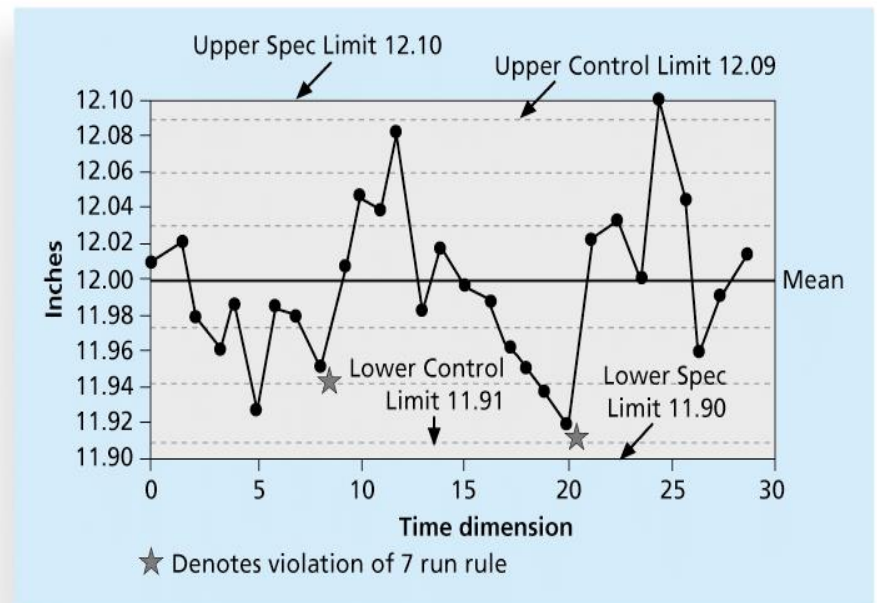
Cause and effect diagrams

- ▶ Cause-and-effect diagrams trace complaints about quality problems back to the responsible production operations
- ▶ They help you find the root cause of a problem
- ▶ Also known as fishbone or Ishikawa diagrams
- ▶ Can also use the 5 whys technique where you repeated ask the question “Why” (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause



Quality Control Charts

- ▶ A **control chart** is a graphic display of data that illustrates the results of a process over time
- ▶ The main use of control charts is to prevent defects, rather than to detect or reject them
- ▶ The **seven run rule** states that if seven data points in a row are all below the mean, above the mean, or are all increasing or decreasing, then the process needs to be examined for non-random problems



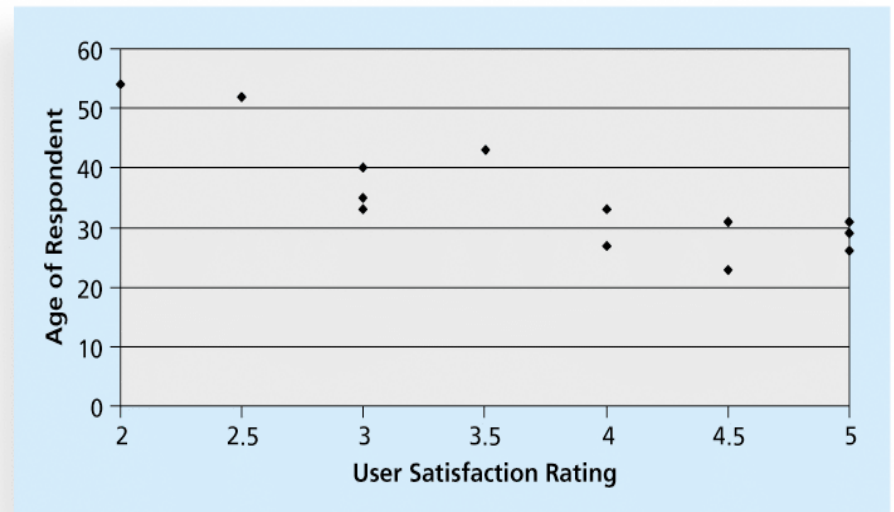
Check sheet

- ▶ A check sheet is used to collect and analyze data
- ▶ It is sometimes called a tally sheet or checklist, depending on its format
- ▶ In the figure, most complaints arrive via text message, and there are more complaints on Monday and Tuesday than on other days of the week
- ▶ This information might be useful in improving the process for handling complaints

System Complaints								
Source	Day							Total
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	
Email								12
Text								29
Phone call								8
Total	11	10	8	6	7	3	4	49

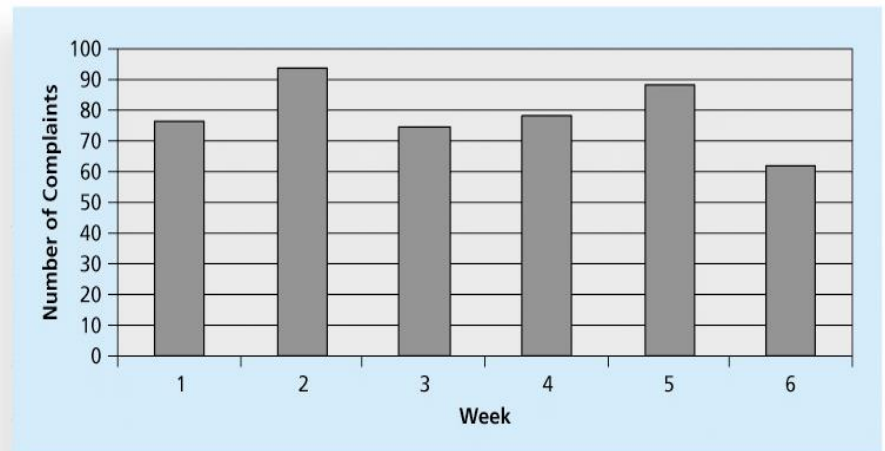
Scatter Diagram

- ▶ A **scatter diagram** helps to show if there is a relationship between two variables
- ▶ The closer data points are to a diagonal line, the more closely the two variables are related



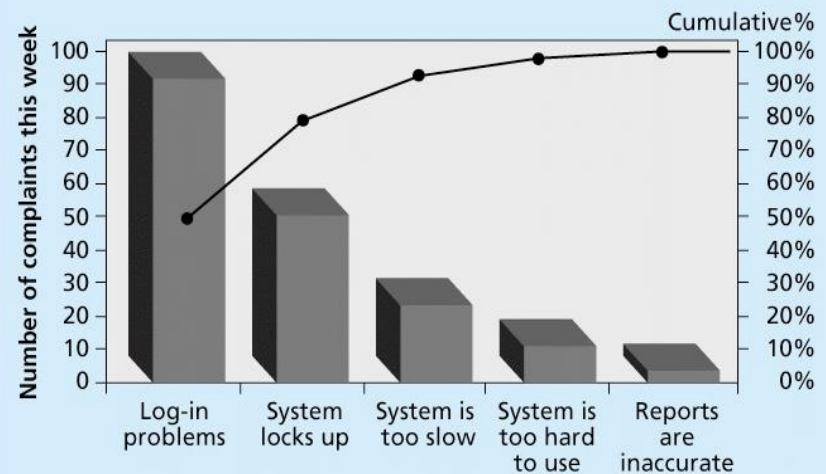
Histograms

- ▶ A **histogram** is a bar graph of a distribution of variables
- ▶ Each bar represents an attribute or characteristic of a problem or situation, and the height of the bar represents its frequency



Pareto Charts

- ▶ A **Pareto chart** is a histogram that can help you identify and prioritize problem areas
- ▶ **Pareto analysis** is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes



Flow Charts

- ▶ Flowcharts are graphic displays of the logic and flow of processes that help you analyze how problems occur and how processes can be improved
- ▶ They show activities, decision points, and the order of how information is processed

